Spark介绍与环境配置



安装Apache Spark



❑ Windows

1. JDK8 or JDK11

2. Python3.6 or higher

3. Hadoop WinUtils

4. Spark Binaries

5. Environment Variables

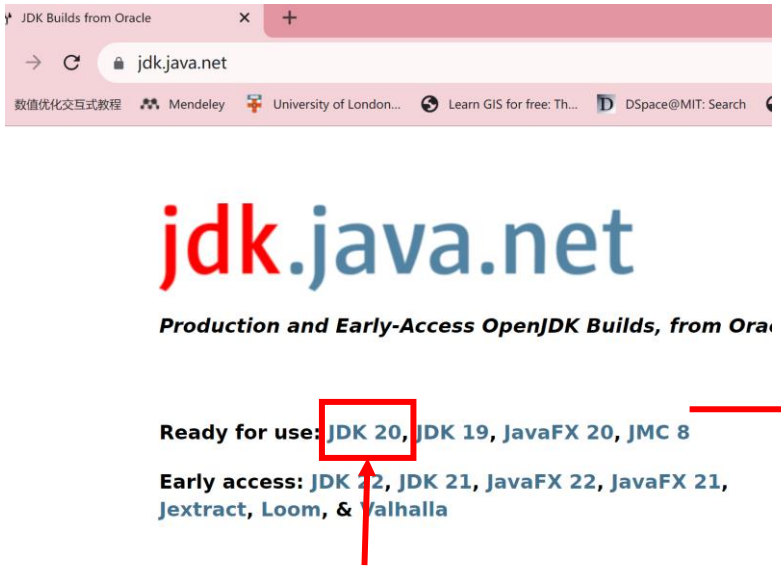
6. Python IDE

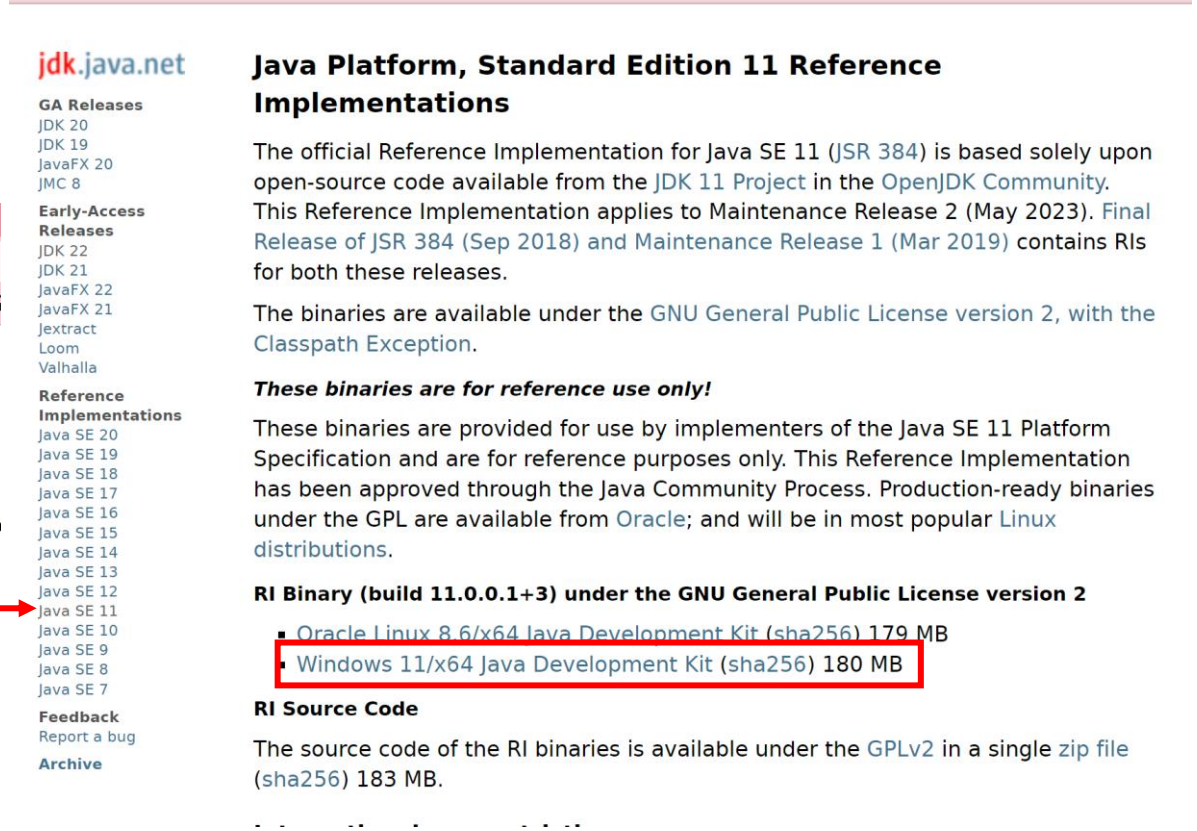
安装Apache Spark

❑ Windows

1. JDK8 or JDK11

搜索：jdk.java.net



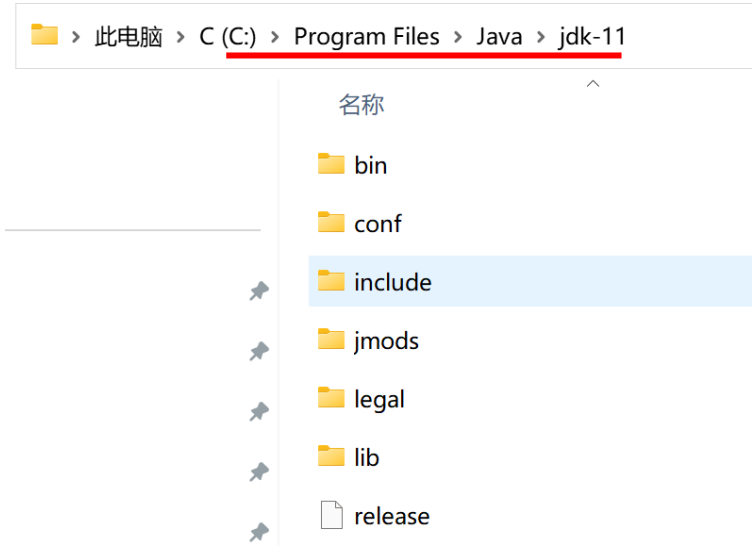


安装Apache Spark

❑ Windows

1. JDK8 or JDK11

下载之后，解压，将其改名为jdk-11,将这个文件夹拷贝到C:\Program Files\Java下



现在有了jdk11，但是还不能运行，需要设置2个环境变量
首先打开command prompt(命令提示符)， 输入：
setx JAVA_HOME "C:\Program Files\Java\jdk-11"

```
C:\Users\yangmin>setx JAVA_HOME "C:\Program Files\Java\jdk-11"

成功: 指定的值已得到保存。
```

```
C:\Users\yangmin>echo %JAVA_HOME%
C:\Program Files\Java\jdk-11
```

查看(需要退出重新进入):
echo %JAVA_HOME%

安装Apache Spark

❑ Windows

1. JDK8 or JDK11

其次，还需要将 JAVA_HOME\bin 添加到PATH环境变量中去

setx PATH "%PATH%;%JAVA_HOME%\bin"

```
C:\Users\yangmin>setx PATH "%PATH%;%JAVA_HOME%\bin"

成功: 指定的值已得到保存。
```

退出后，测试是否安装成功:
java -version

```
C:\Users\yangmin>java -version
openjdk version "11.0.0.1" 2023-05-09
OpenJDK Runtime Environment 18.9 (build 11.0.0.1+3-5)
OpenJDK 64-Bit Server VM 18.9 (build 11.0.0.1+3-5, mixed mode)
```

2. Python3.6 or higher

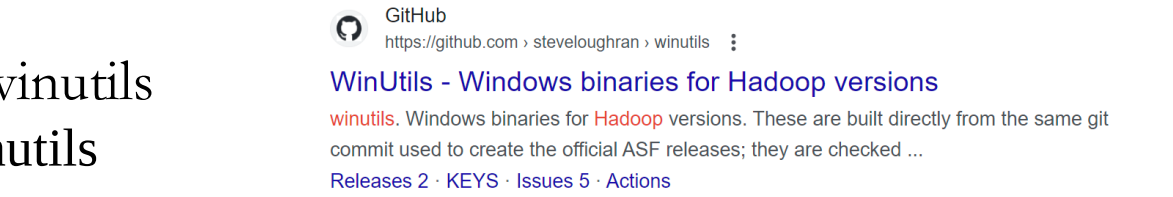
```
C:\Users\yangmin>python --version
Python 3.10.9
```

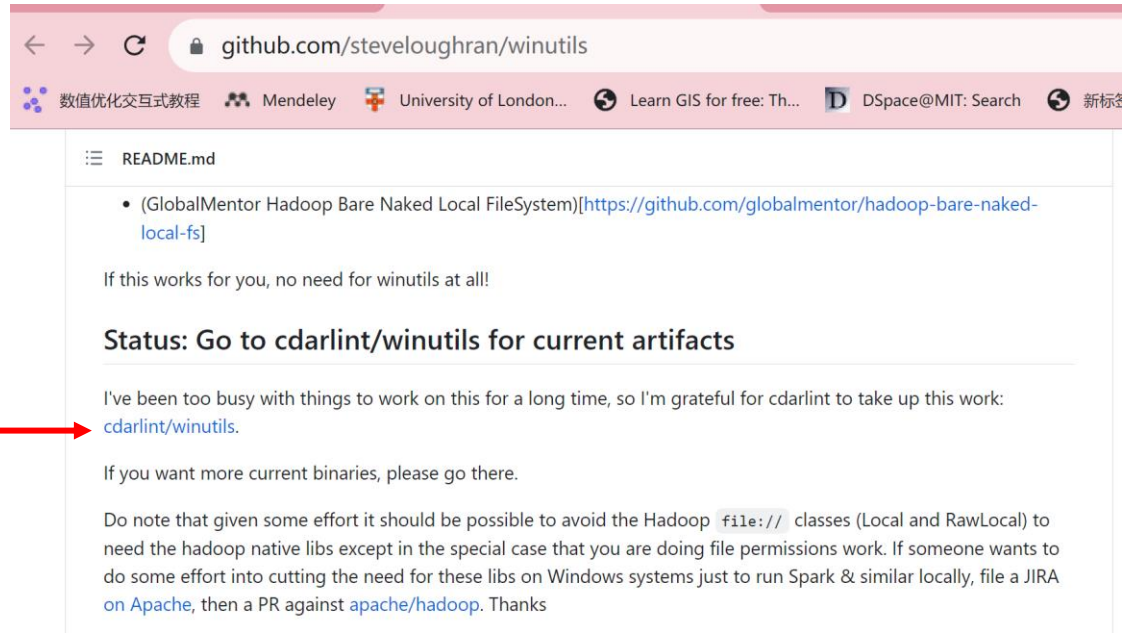
安装Apache Spark

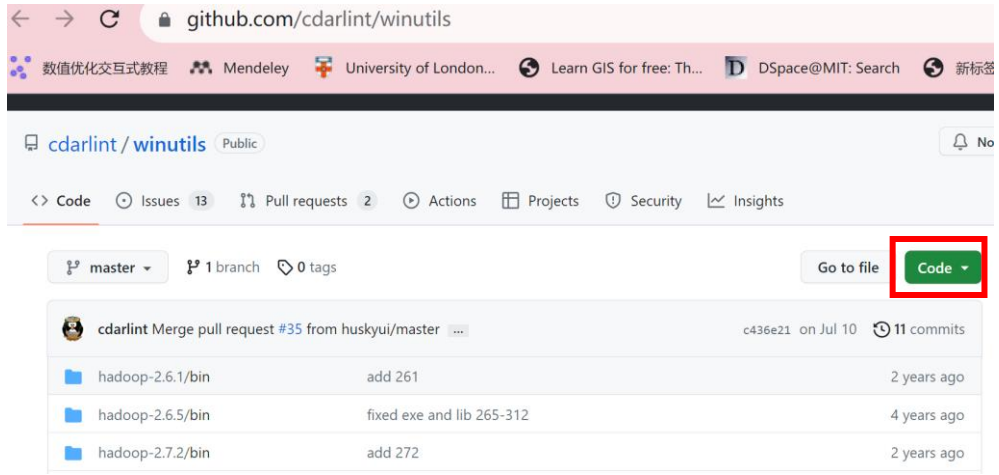
❑ Windows

3. 1 安装Hadoop winutils

搜索： Hadoop winutils







下载并解压

how to use

这个页面下方有如何使用：

place a copy of hadoop-ver folder on your local drive set environment vars:

```
HADOOP_HOME=<your local hadoop-ver folder>
PATH=%PATH%;%HADOOP_HOME%\bin
```

需要设置两个环境变量

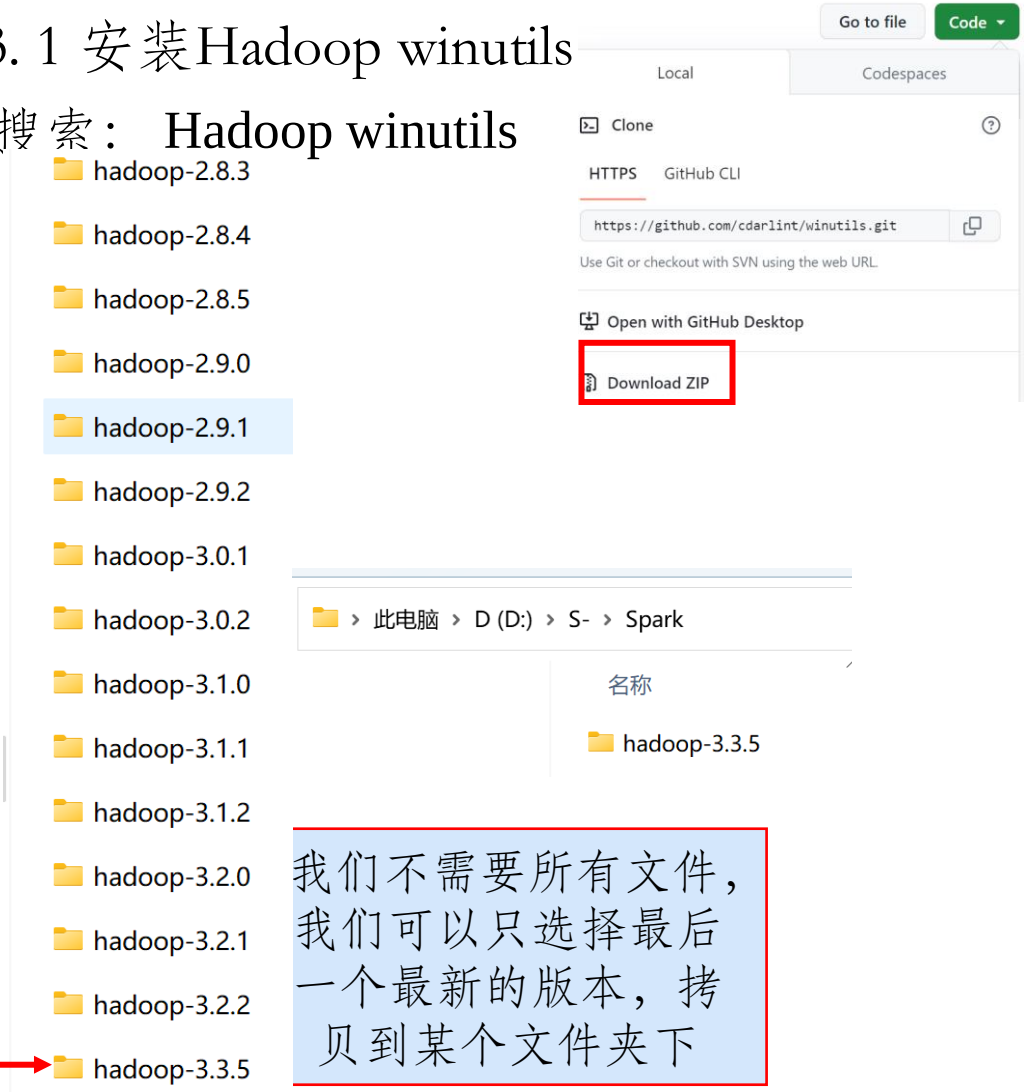
then you'll pass the "no native library" and "access0" error

安装Apache Spark

❑ Windows

3. 1 安装Hadoop winutils

搜索： Hadoop winutils



我们不需要所有文件，我们可以只选择最后一个最新的版本，拷贝到某个文件夹下

打开命令提示符：

setx HADOOP_HOME "D:\S-\Spark\hadoop-3.3.5"

```
C:\Users\yangmin>setx HADOOP_HOME "D:\S-\Spark\hadoop-3.3.5"

成功： 指定的值已得到保存。
```

指定路径：

setx PATH "%PATH%;%HADOOP_HOME%\bin"

```
C:\Users\yangmin>setx PATH "%PATH%;%HADOOP_HOME%\bin"

成功： 指定的值已得到保存。
```

安装Apache Spark

- Windows

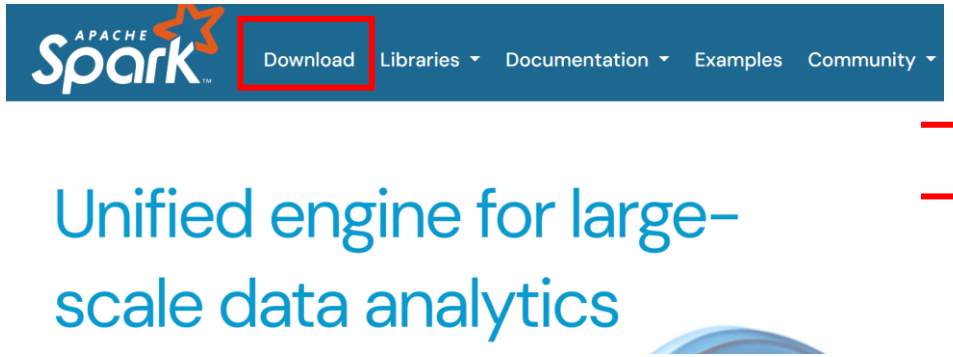
3. 2 安装Hadoop

安装Apache Spark

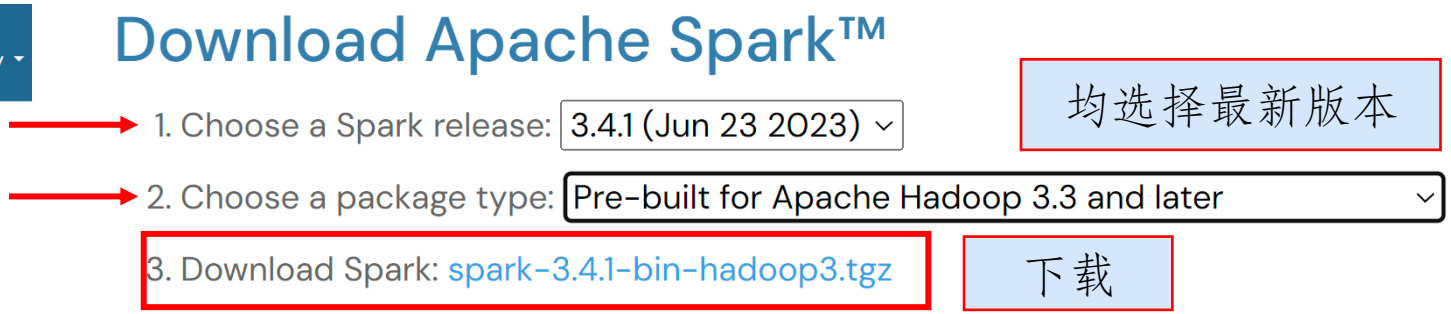
❑ Windows

4. Spark Binaries

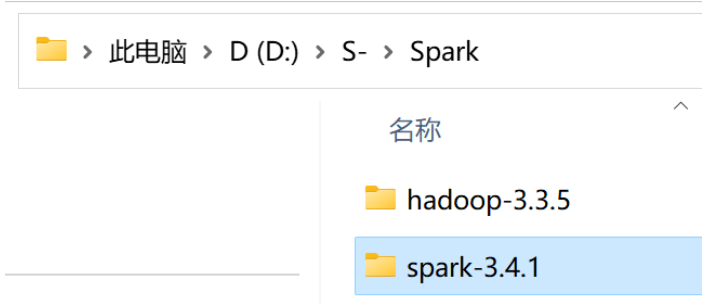
搜索： apache spark







下载，解压之后拷贝到某个文件夹下（可以改名）



安装Apache Spark

❑ Windows

4. Spark Binaries

命令提示符：设置环境变量

setx SPARK_HOME "D:\S-\Spark\spark-3.4.1 “

setx PATH "%PATH%;%SPARK_HOME%\bin"

```
C:\Users\yangmin>setx SPARK_HOME "D:\S-\Spark\spark-3.4.1"

成功: 指定的值已得到保存。

C:\Users\yangmin>setx PATH "%PATH%;%SPARK_HOME%\bin"

警告: 正保存的数据被裁断到 1024 字符。

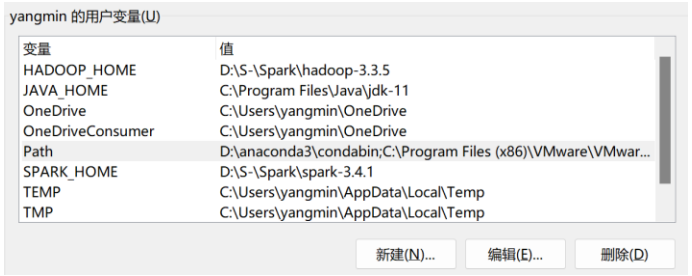
成功: 指定的值已得到保存。
```

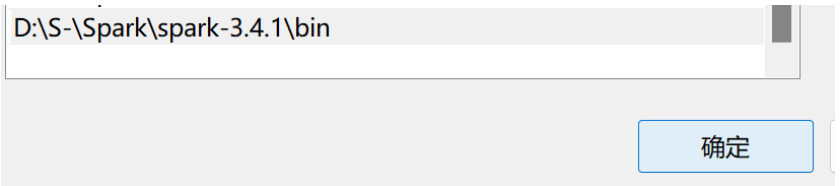
此警告出现在某些 PATH 环境变量太长的 Windows 计算机上,并且您不能使用 setx 命令向 PATH 添加更多内容.但是,您可以使用 Windows UI 添加更扩展的 PATH 环境变量

具体操作：

我的电脑（此电脑）右键-属性-高级系统设置-环境变量-PATH-编辑-新建—输入：D:\S-\Spark\spark-3.4.1\bin 确定即可







安装Apache Spark

❑ Windows

4. Spark Binaries

测试是否安装成功，命令提示符

pyspark

```
C:\Users\yangmin>pyspark
Python 3.10.9 | packaged by Anaconda, Inc. | (main, Mar  1 2023, 18:18:15) [MSC v.1916 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
23/08/26 12:33:41 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java cl
asses where applicable
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.4.1
      /_/

Using Python version 3.10.9 (main, Mar  1 2023 18:18:15)
Spark context Web UI available at http://min:4040
Spark context available as 'sc' (master = local[*], app id = local-1693024423020).
SparkSession available as 'spark'.
>>>
```

1.Python Path
2. Pyspark_Python

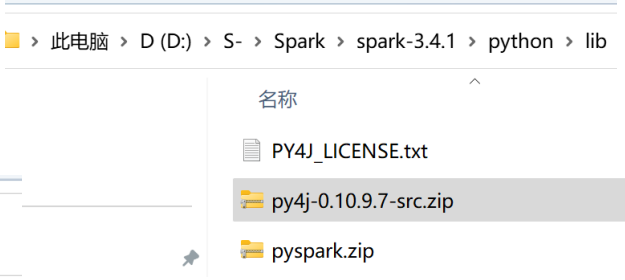
如果在运行 PySpark shell 时看到任何错误，可能需要设置以下环境变量。
建议设置这些环境变量即使您在运行 pyspark shell 时没有看到任何错误。

安装Apache Spark

❑ Windows

5. Environment Variables





设置 Python Path

$ setx PYTHONPATH "D:\S-\Spark\spark-3.4.1\python; D:\S-\Spark\spark-3.4.1\python\lib\py4j-0.10.9.7-src.zip"

```
C:\Users\yangmin>setx PYTHONPATH "D:\S-\Spark\spark-3.4.1\python; D:\S-\Spark\spark-3.4.1\python\lib\py4j-0.10.9.7-src.zip"

成功: 指定的值已得到保存。
```

设置 Pyspark_Python

$ where python

```
C:\Users\yangmin>where python
D:\anaconda3\python.exe
C:\Users\yangmin\AppData\Local\Microsoft\WindowsApps\python.exe
```

$ setx PYSPARK_PYTHON "D:\anaconda3\python.exe"

```
C:\Users\yangmin>setx PYSPARK_PYTHON "D:\anaconda3\python.exe"

成功: 指定的值已得到保存。
```

测试

$ pyspark

```
C:\Users\yangmin>pyspark
Python 3.10.9 | packaged by Anaconda, Inc. | (
Type "help", "copyright", "credits" or "licens
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(new
23/08/26 14:07:07 WARN NativeCodeLoader: Unabl
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.4.1
      /_/

Using Python version 3.10.9 (main, Mar  1 2023
Spark context Web UI available at http://min:4
Spark context available as 'sc' (master = loca
SparkSession available as 'spark'.
>>>
```

安装Apache Spark

❑ Windows

6. Python IDE

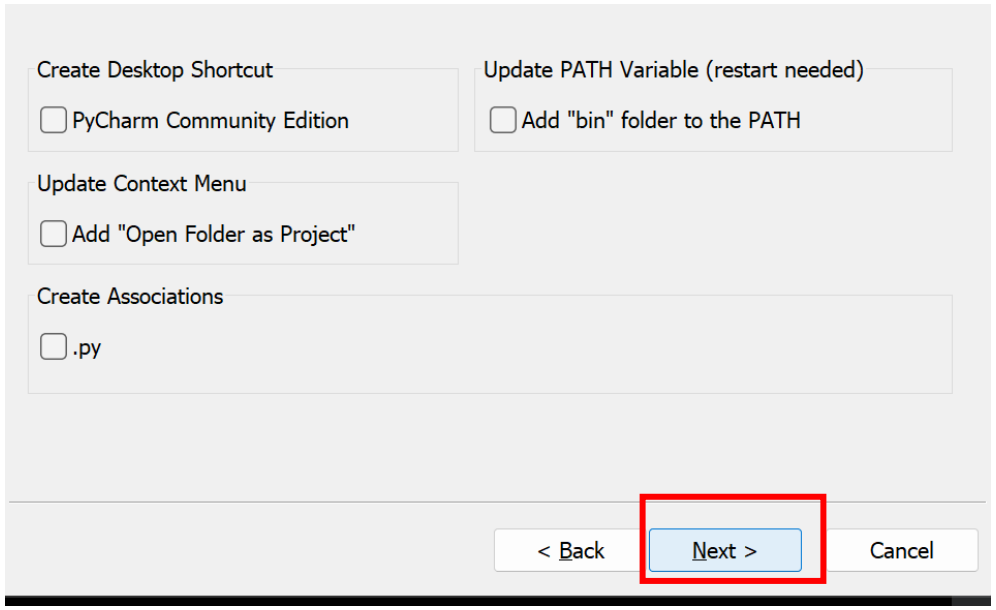
搜索pycharm

JetBrains
https://www.jetbrains.com › pycharm

PyCharm: the Python IDE for Professional Developers by ...

The Python & Django IDE with intelligent code completion, on-the-fly error checking, quick-fixes, and much more...

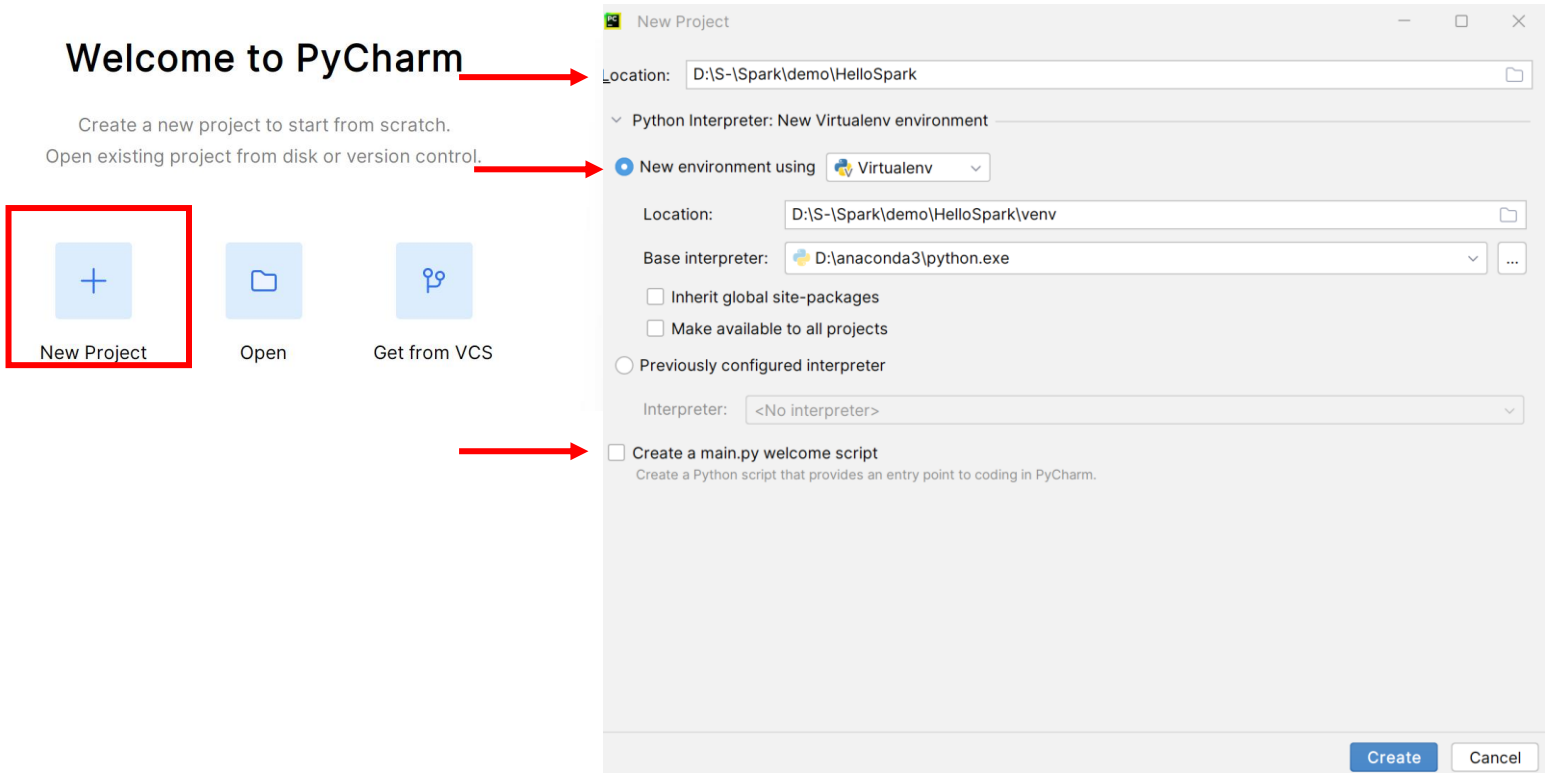




安装 -
全部默认

安装Apache Spark

❑ Windows下创建第一个Spark Application



新建的project下再新建一个python文件

安装Apache Spark

- 使用Python IDE创建第一个Spark Application

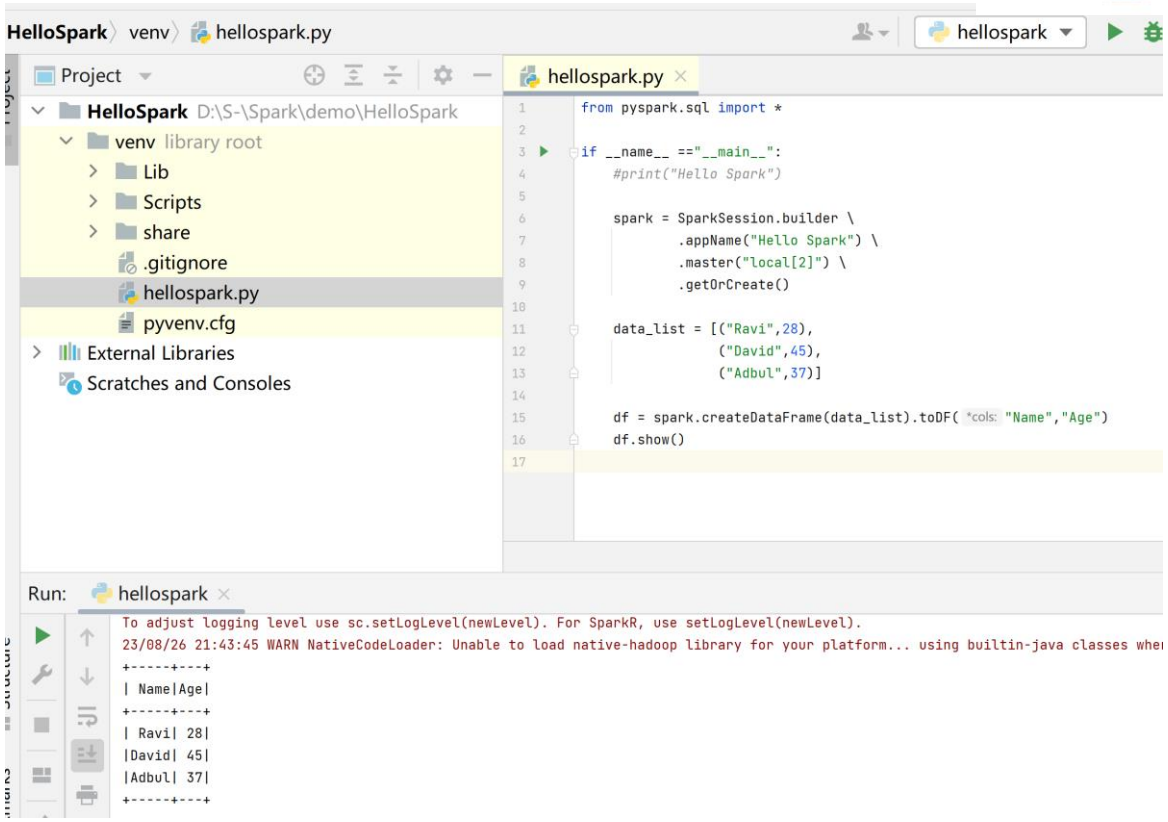
```
from pyspark.sql import *

if __name__ =="__main__":
    #print("Hello Spark")

    spark = SparkSession.builder \
            .appName("Hello Spark") \
            .master("local[2]") \
            .getOrCreate()

    data_list = [("Ravi",28),
                 ("David",45),
                 ("Adbul",37)]

    df =
spark.createDataFrame(data_list).toDF("Name","Age")
    df.show()
```



安装Apache Spark



- Windows

Spark集群安装：

Java, Hadoop, Scala, Spark

安装Apache Spark

❑ Linux

1. 安装VMware

2. 安装Ubuntu 镜像文件 ios:

打开vm,选择创建新的虚拟机---> 自定义（下一步）下一步 ---> 安装程序光盘映像文件（ubuntu 18.04.3）

安装位置：D:\S-\Spark-Linux\clusters_ubuntu18\master

处理器数量：2，每个处理器的内核数量：2； 内存：8192MB； 网络：NAT

3. 使用命令调整界面大小

$ xrandr # 当前显示窗口大小

$ xrandr -s 1280x960 # 注意：中间不是乘号，是字母x

安装Apache Spark

❑ Linux

4. 安装Spark环境

4.1 可能需要的操作：更新apt，安装vim

$ sudo apt-get update

$ sudo apt install net-tools

$ sudo apt-get install vim

$ sudo apt install rpm

可能出现的问题：windows和terminal无法复制粘贴

https://blog.csdn.net/YZL40514131/article/details/122281396

安装Apache Spark

❑ Linux

4.2 安装java

（1）下载https://www.oracle.com/java/technologies/downloads/

（2）创建目录来保存JDK

```
$ cd /usr/lib
$ sudo mkdir jvm
```

（3）解压安装

```
$ cd ~/Downloads
$ sudo tar -zxvf ./jdk-8u381-linux-x64.tar.gz -C /usr/lib/jvm
```

（4）配置java环境

```
$ sudo gedit ~/.bashrc
```

```
export JAVA_HOME=/usr/lib/jvm/jdk1.8.0_381
export JRE_HOME=${JAVA_HOME}/jre
export CLASSPATH=.:${JAVA_HOME}/lib:${JRE_HOME}/lib
export PATH=${JAVA_HOME}/bin:$PATH
```

```
# 使环境生效:
$ source ~/.bashrc

# 检验安装成功?
$ java -version
```

安装Apache Spark

❑ Linux

4.3 安装Scala

（1）下载好之后粘贴到 home/Downloads下

（2）+（3）创建目录保存，解压

$ sudo tar -xvf scala-2.12.2.tgz -C <设定目录>

（4）环境配置

$ sudo gedit ~/.bashrc

export SCALA_HOME=/home/min//scala-2.12.11

export PATH=$SCALA_HOME/bin:$PATH

$使环境生效：
$ source ~/.bashrc
检验安装成功?
$ scala-version

安装Apache Spark

❑ Linux

4.4 安装Hadoop

（1）配置环境文件之后，需要注意的是，需要修改一些文件：

/etc/hadoop 下的： core-site.xml　hdfs-site.xml　hadoop-env.sh

（2）首次部署配置文件之后，需要对文件系统进行格式化

cd到 hadoop/bin:

$ hdfs namenode – format

```
/************************************************************
SHUTDOWN_MSG: Shutting down NameNode at master/192.168.233.140
************************************************************/
```

（3）运行之前可以做清空处理

cd到 hadoop:

$ rm -rf tmp

$ rm – rf logs/*

安装Apache Spark

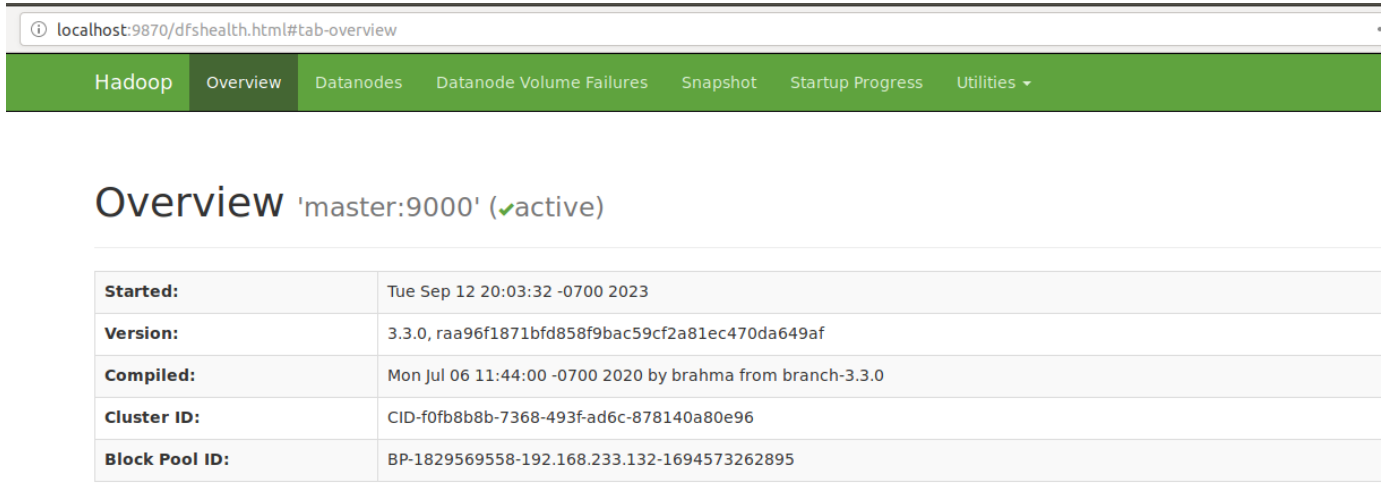
❑ Linux

4.4 安装Hadoop

（4）hadoop/sbin 下：

$ start-dfs.sh

```
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [master]
```



安装Apache Spark

❑ Linux

4.5 安装Spark

https://archive.apache.org/dist/spark

```
min@master:~$ spark-shell
23/09/08 00:14:56 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java class
es where applicable
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Spark context Web UI available at http://master:4040
Spark context available as 'sc' (master = local[*], app id = local-1694157301268).
Spark session available as 'spark'.
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 3.0.0
      /_/

Using Scala version 2.12.10 (Java HotSpot(TM) 64-Bit Server VM, Java 1.8.0_381)
Type in expressions to have them evaluated.
Type :help for more information.
```

安装Apache Spark

❑ Linux

5. 配置集群环境

参考：https://blog.csdn.net/m0_61022929/article/details/126403124

（1）修改主机名（每台） /etc/hostname 改为： master, slave1, slave2

（2）设置master和slave的ip /etc/hosts

（3）全局环境变量配置： /etc/profile

hadoop和spark的环境变量配置

（4）ssh无密码登录

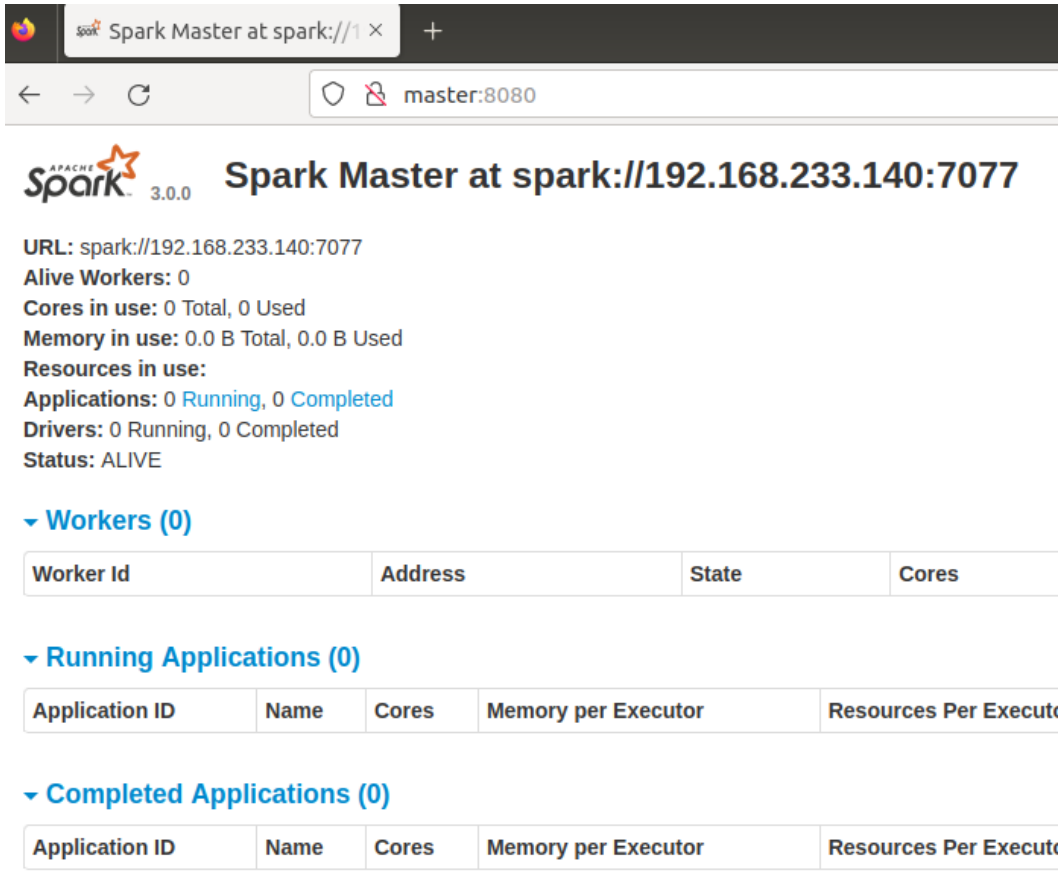
安装Apache Spark

❑ Linux

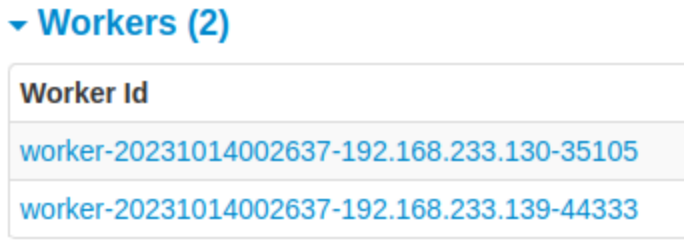
5. 配置集群环境

（5）启动集群　spark/sbin $ sh start-all.sh

```
starting org.apache.spark.deploy.master.Master, logging
-1-master.out
slave2: starting org.apache.spark.deploy.worker.Worker,
r.Worker-1-slave2.out
slave1: starting org.apache.spark.deploy.worker.Worker,
r.Worker-1-slave1.out
```



如果没有works：解决办法：
1. 关闭防火墙
systemctl status firewalld# 查看防火墙状态
systemctl stop firewalld# 暂时关闭防火墙
systemctl disable firewalld#永久关闭
2. slave下: spark/sbin/ $ sh start-slave.sh spark://



Why Spark?

❑ Big Data

What is Big Data?

Big data refers to extremely large and diverse collections of structured, unstructured, and semi-structured data that continues to grow exponentially over time. These datasets are so huge and complex in volume, velocity, and variety, that traditional data management systems cannot store, process, and analyze them.

https://cloud.google.com/learn/what-is-big-data

The Google File System

Sanjay Ghemawat, Howard Gobioff, and Shun-Tak Leung

Google*



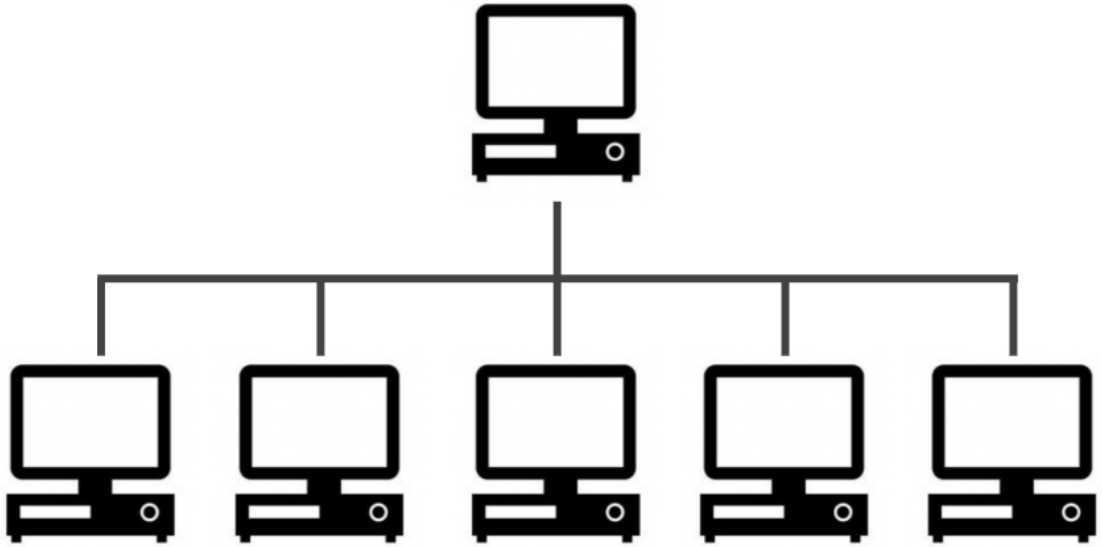
Why Spark?

- Big Data

单机系统和分布式系统



Local

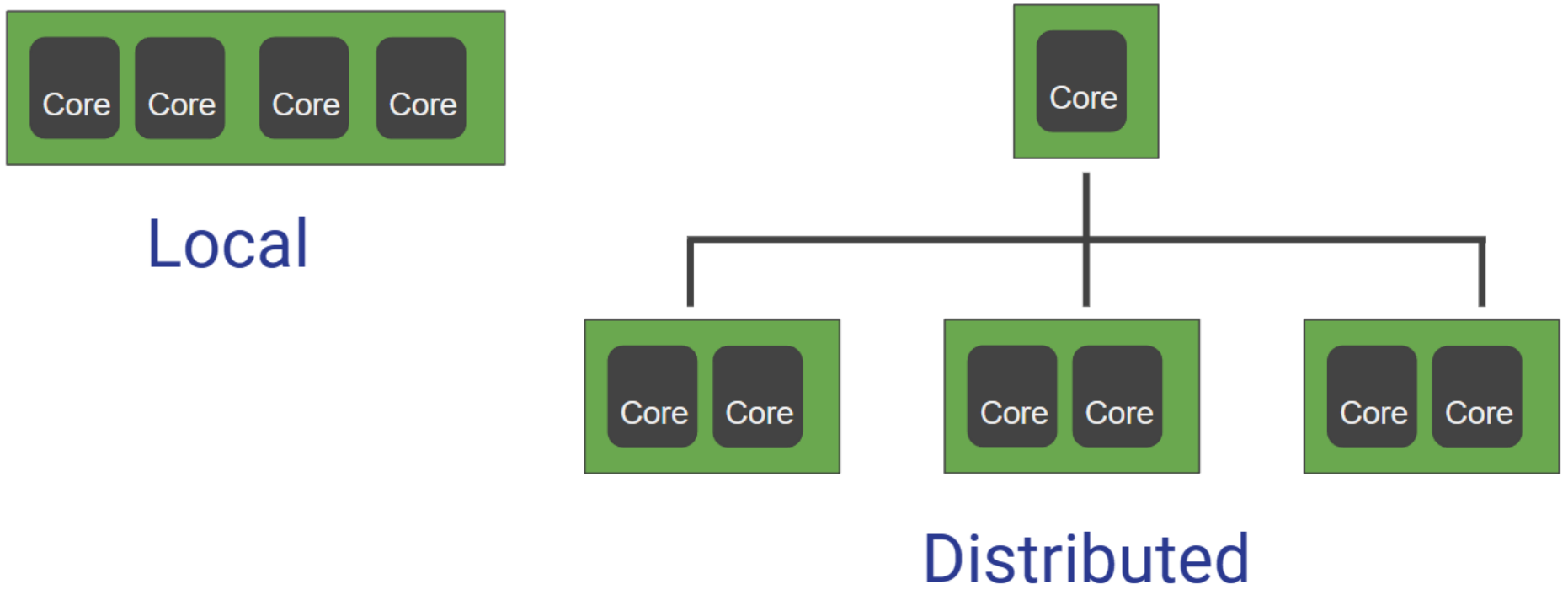


Distributed

Why Spark?

- Big Data

单机系统和分布式系统



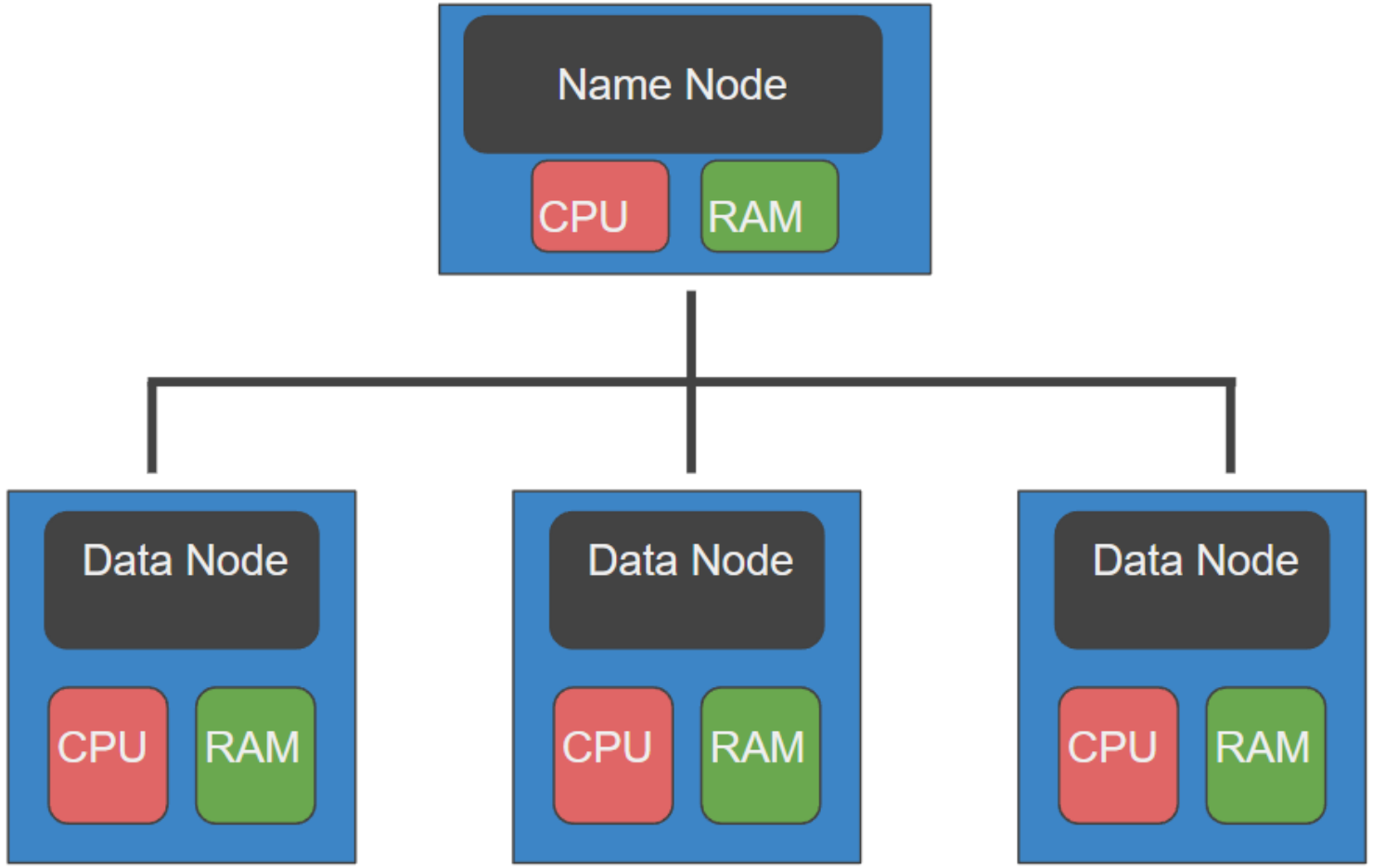
Why Spark?

❑ Hadoop

- Hadoop is a way to distribute very large files across multiple machines.
- It uses the Hadoop Distributed File System (HDFS)
- HDFS allows a user to work with large data sets
- HDFS also duplicates blocks of data for fault tolerance
- It also then uses MapReduce
- MapReduce allows computations on that data

Why Spark?

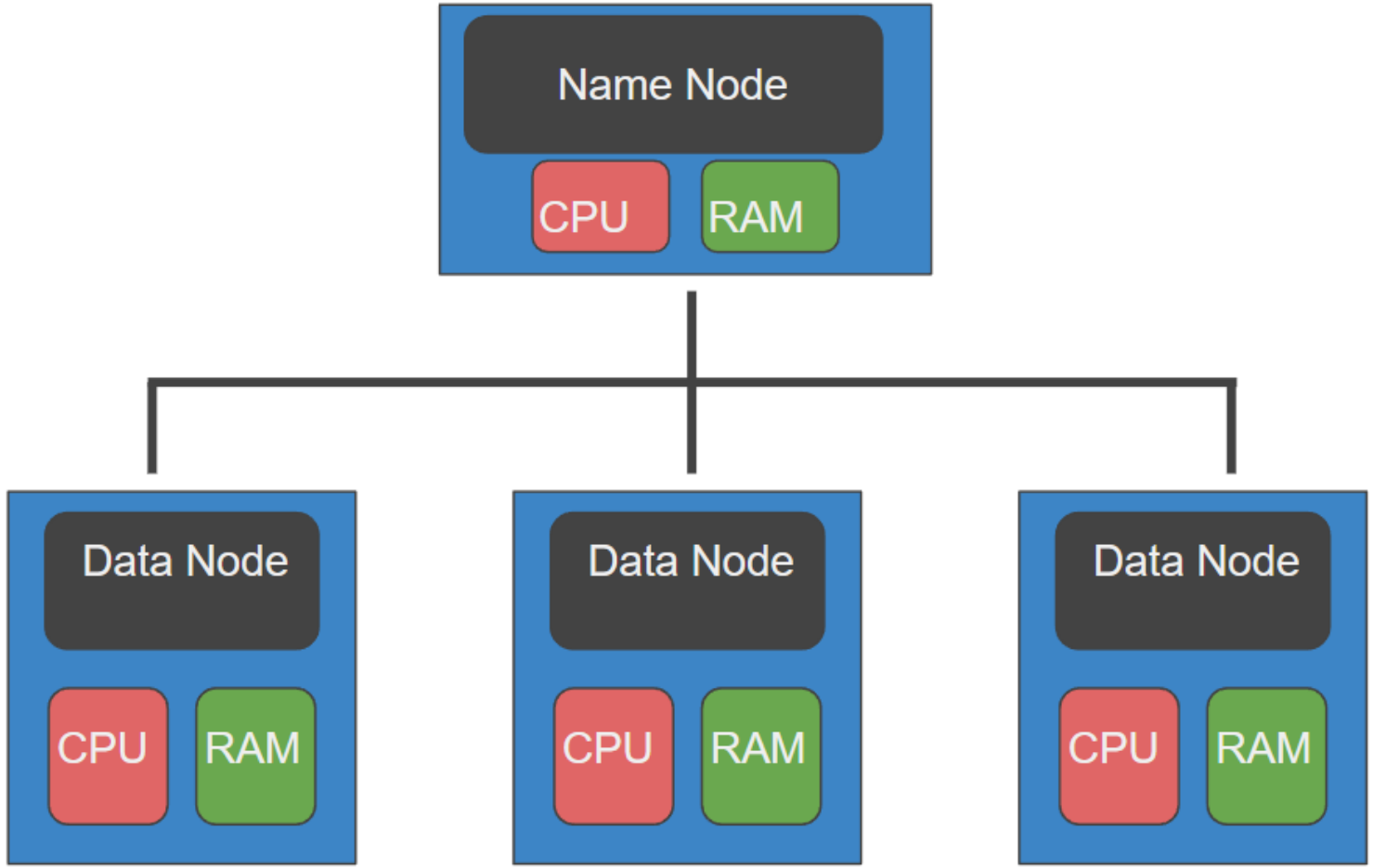
- Hadoop - HDFS



Why Spark?

❑ Hadoop - HDFS

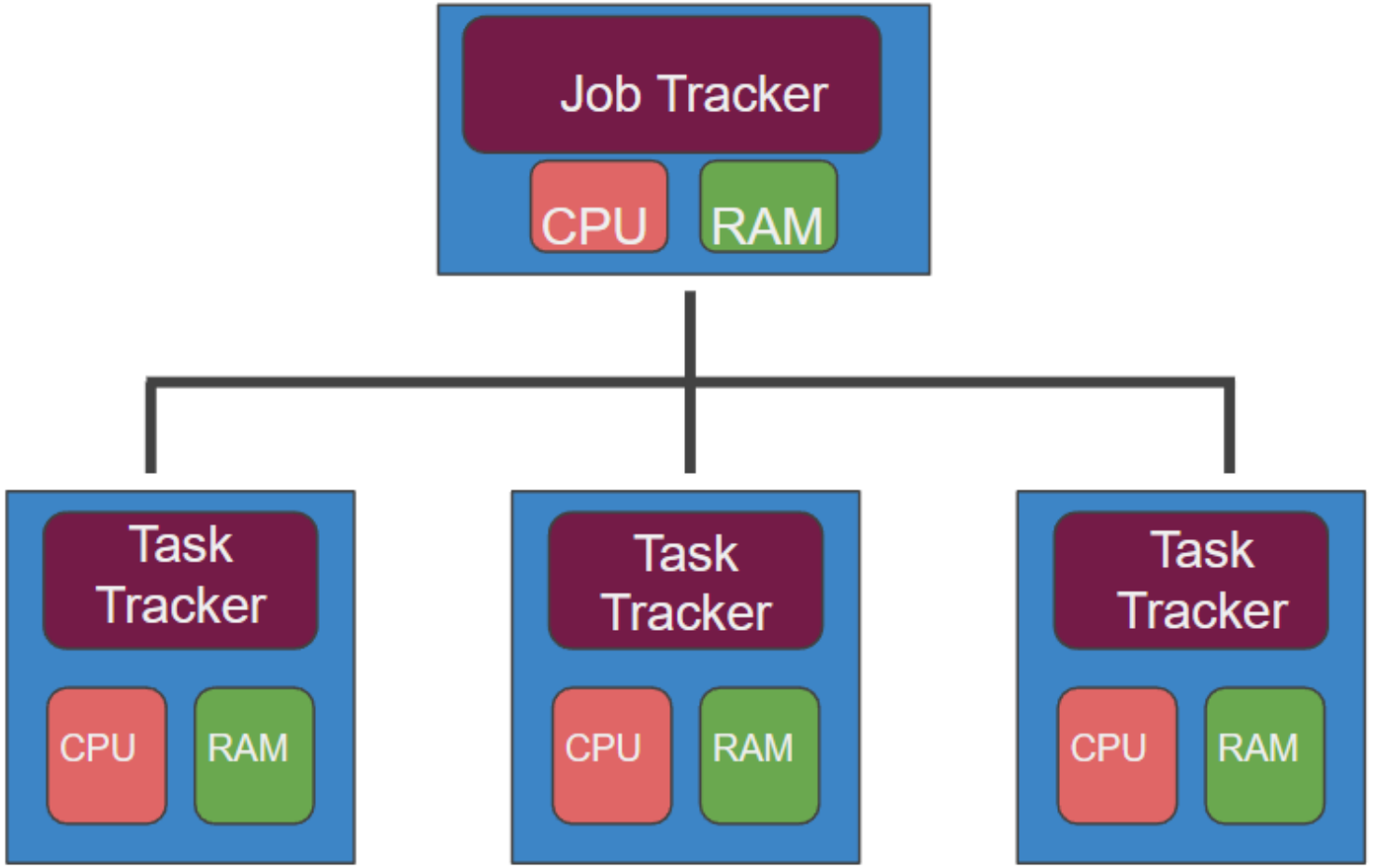
- HDFS will use blocks of data, with a size of 128 MB by default
- Each of these blocks is replicated 3 times
- The blocks are distributed in a way to support fault tolerance
- Smaller blocks provide more parallelization during processing
- Multiple copies of a block prevent loss of data due to a failure of a node



Why Spark?

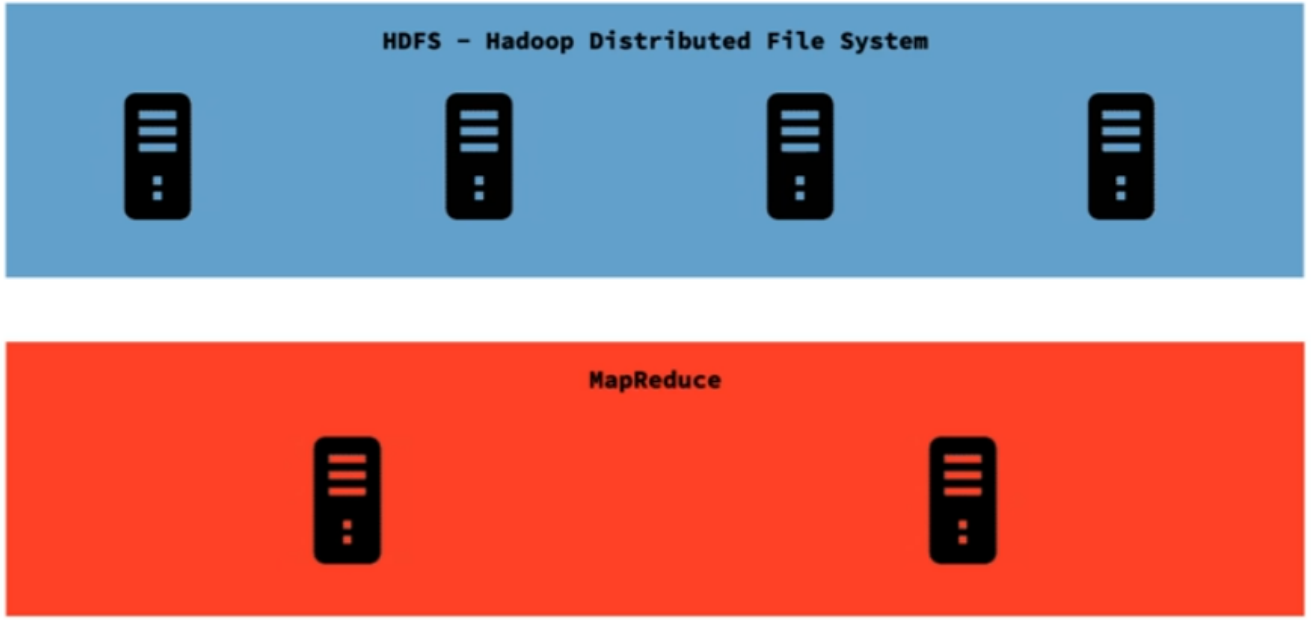
❑ Hadoop - MapReduce

- MapReduce is a way of splitting a computation task to a distributed set of files (such as HDFS)
- It consists of a Job Tracker and multiple Task Trackers
- The Job Tracker sends code to run on the Task Trackers
- The Task trackers allocate CPU and memory for the tasks and monitor the tasks on the worker nodes



Why Spark?

- Big Data - Hadoop



The Google File System

Sanjay Ghemawat, Howard Gobioff, and Shun-Tak Leung

Google*



Why Spark?

❑ Big Data – Apache Spark



◆Apache Spark 是MapReduce的增强版

◆尽管Spark相对于Hadoop而言具有较大优势，但Spark并不能完全替代Hadoop，Spark主要用于替代Hadoop中的MapReduce计算模型。存储依然可以使用HDFS，但是中间结果可以存放在内存中；调度可以使用Spark内置的，也可以使用更成熟的调度系统YARN等。

◆课程内容：

- Spark
- Spark VS MapReduce
- Spark RDDs
- Spark DataFrames